Case Study 1: University Scholarship Allocation

Aim: To develop a Python program that evaluates university students for scholarships based on aggregate marks, subject-wise minimum marks, attendance percentage, and annual family income. The program identifies eligible students, ranks them, and resolves ties using predefined rules.

1. Computational Thinking

Input:

- Student name
- Aggregate marks
- Subject-wise marks
- Attendance percentage
- Annual family income

Processing:

1. Accept student details.
2. Verify minimum subject marks.
3. Check aggregate marks.
4. Check attendance percentage.
5. Check family income.
6. Separate eligible and ineligible students.
7. Rank eligible students based on predefined rules.

Output:

- Eligibility status.
- Ranked list of eligible students.
- Total number of eligible students.

2. Algorithm

1. Start.
2. Read the number of applicants.

3. Create an empty list to store eligible students.
4. Read each student's details.
5. Check whether every subject mark is at least 40.
6. Check whether aggregate marks are at least 75.
7. Check whether attendance is at least 75%.
8. Check whether annual family income is at most ₹2,50,000.
9. If all conditions are satisfied, add the student to the eligible list.
10. Sort eligible students using the following rules:
    - Higher aggregate marks first.
    - Higher attendance first.
    - Lower family income first.
11. Display student rankings.
12. Display the total number of eligible students.
13. Stop.

3. Flowchart

Define the following terms: Computer programming Programming langua..

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4. Pseudocode

START

INPUT n

CREATE empty list students

FOR i = 1 TO n

```
INPUT name
INPUT aggregate
INPUT subject marks
INPUT attendance
INPUT income

SET subject_eligible = TRUE

FOR each subject mark

    IF subject mark < 40
        SET subject_eligible = FALSE
    END IF

END FOR

IF aggregate >= 75 AND
   subject_eligible = TRUE AND
   attendance >= 75 AND
   income <= 250000

   ADD student to students

ELSE

   DISPLAY "Not Eligible"
```

END IF

END FOR

SORT students BY

aggregate DESCENDING

attendance DESCENDING

income ASCENDING

DISPLAY ranked students

STOP

5. Complete Python Program

```
students = []

n = int(input("Enter number of applicants: "))

for i in range(n):

    print("\nEnter details of Student", i + 1)

    name = input("Enter student name: ")

    aggregate = float(input("Enter aggregate marks: "))
```

```
subjects = []

print("Enter marks of 3 subjects:")

for j in range(3):
    marks = float(input("Enter subject marks: "))
    subjects.append(marks)

attendance = float(input("Enter attendance percentage: "))

income = float(input("Enter annual family income: "))

# Check subject-wise eligibility

subject_eligible = True

for marks in subjects:

    if marks < 40:
        subject_eligible = False

# Check overall eligibility

if (aggregate >= 75 and
    subject_eligible and
    attendance >= 75 and
```

```
        income <= 250000):

        student = (name, aggregate, attendance, income)

        students.append(student)

        print(name, "is Eligible")

    else:

        print(name, "is Not Eligible")



# Rank eligible students

students.sort(key=lambda x: (-x[1], -x[2], x[3]))



# Display scholarship results

print("\n========== SCHOLARSHIP RESULTS ==========")

if len(students) == 0:

    print("No eligible students found.")
```

```
else:

    print("\nRank\tName\tAggregate\tAttendance\tIncome")

    rank = 1

    for student in students:

        name, aggregate, attendance, income = student

        print(rank, "\t", name, "\t",
              aggregate, "\t\t",
              attendance, "\t\t", income)

        rank = rank + 1


print("\nTotal Eligible Students:", len(students))

print("Scholarship Allocation Completed.")
```

6. Sample Input

Enter number of applicants: 3

Enter details of Student 1

Enter student name: Ravi

Enter aggregate marks: 85

Enter marks of 3 subjects:

Enter subject marks: 80

Enter subject marks: 90

Enter subject marks: 85

Enter attendance percentage: 90

Enter annual family income: 200000

Enter details of Student 2

Enter student name: Sita

Enter aggregate marks: 90

Enter marks of 3 subjects:

Enter subject marks: 95

Enter subject marks: 85

Enter subject marks: 90

Enter attendance percentage: 95

Enter annual family income: 180000

Enter details of Student 3

Enter student name: Arun

Enter aggregate marks: 70

Enter marks of 3 subjects:

Enter subject marks: 80

Enter subject marks: 75

Enter subject marks: 70

Enter attendance percentage: 85

Enter annual family income: 150000

7. Expected Output

Ravi is Eligible

Sita is Eligible

Arun is Not Eligible

========== SCHOLARSHIP RESULTS ==========

Rank Name Aggregate Attendance Income

1 Sita 90 95 180000

2 Ravi 85 90 200000

Total Eligible Students: 2

Scholarship Allocation Completed.

8. Syllabus Concepts Used

Syllabus topic	**Application**
Computational thinking	Breaking the problem into steps
Variables	name, aggregate, attendance
Data types	int, float, string

| Operators | >=, Case Study 1 completed.